

A Sandbox Study of 3D YOLO and Anomaly Detection

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Abstract:

This paper presents a simplified 3D YOLO anomaly detection framework.

Introduction:

Our framework focuses on spatial anomaly detection in industrial environments.

It uses a 3D YOLO-based architecture with real-time video processing.

However, it has one major limitation that affects real-world deployment.

Method:

We employ a custom 3D convolutional backbone with YOLO-style detection heads.

The model is trained on synthetic anomaly data with mixed results.

Conclusion:

The limitation is that the model requires high lighting conditions to avoid false alarms. Future work should address this constraint.

References:

[1] Redmon et al., YOLOv3, 2018.

[2] Doe et al., 3D Convolutions for Video, 2020.